Title of the project:

Internal Intrusion Detection System using System Call Analysis

Abstract:

As per today's business needs, protecting digital resources is very important regarding both outside cyber-attacks and misuse by company people. As the existing project system was developed to detect DDoS attacks on the SDN dataset using deep learning models. The work was focused regarding building an effective detection system for network security threats. However, this solution actually worked only for network protection and definitely could not find attacks from real users with proper login details.

To overcome the problem, I proposed Internal Intrusion Detection and Protection System (IIDS) surely focus on finding insider threats. Moreover, this system can detect attacks from people working inside the organization. IIDS does not use only user IDs and passwords like traditional methods. It further monitors user activities continuously at system call level and creates personalized behavioural profiles using data mining and machine learning algorithm itself. The system actually compares current user actions with saved profiles to check if the logged-in person is real or fake.

The system offers significant benefits to multiple stakeholders. Sensitive information and resources are better protected by organizations from both internal and external threats. Strong tools are available to system administrators for the monitoring, identification, and forensic examination of questionable activity in real time. Better account security and a lower chance of identity theft benefit end users. This project offers a thorough framework to protect contemporary computing environments by expanding security from network-level DDoS detection to user-level insider threat avoidance

Keywords: *Internal Intrusion Detection, Insider Threats, System Call Analysis, Behavioural Profiling,*

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